

ROBO TUG OF WAR

INTERNATIONAL ROBOTICS COMPETITION TECHNICAL MANUAL & MATCH RULES

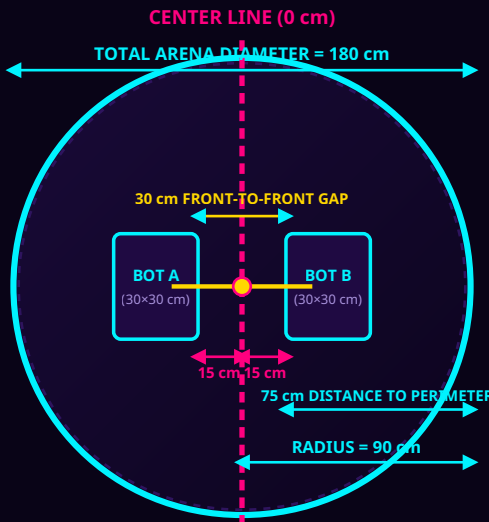
1. CATEGORY TECHNICAL SPECIFICATIONS

All participating teams must comply with official bot dimensions, mass limits (including the $\pm 5\%$ tolerance), and power constraints prior to match qualification.

PARAMETER	JUNIOR CATEGORY (CLASS 6 – 9)	SENIOR CATEGORY (CLASS 10 – 12)	TOLERANCE RULES
Nominal Weight Limit	3.00 kg	5.00 kg	Base weight requirement
Weight Tolerance Range ($\pm 5\%$)	2.85 kg — 3.15 kg	4.75 kg — 5.25 kg	$\pm 5\%$ Allowed Margin
Maximum Dimensions	30 cm × 30 cm × 30 cm	30 cm × 30 cm × 30 cm	Measured in uncompressed state
Max Battery Voltage	12.0 V DC (Nominal)	12.0 V DC (Nominal)	Max 12.6V fully charged LiPo
Control Mechanism	Wired / Wireless (RF / BT)	Wired / Wireless (RF / BT)	Wire slack must be maintained
Tether Hitch Point	Max Height: 10 cm from floor	Max Height: 10 cm from floor	Measured to tow ring center

2. ARENA DIMENSIONS & ROBOT STARTING PLACEMENT

The arena consists of a circular high-traction mat with a total diameter of 180 cm (\$R = 90\$ ext{ cm}\$). The diagram below illustrates exact centimeter measurements from the zero-center line.



Placement Rules & Spatial Metrics:

- Arena Outer Boundary: Circle with 180 cm Diameter (Radius = 90 cm from center).
- Bot Position From Center: Front-most edge of each bot must start exactly 15 cm from center line.
- Inter-Robot Distance: Total gap between bot front faces is 30 cm.
- Distance to Perimeter: Front face of each robot to the outer ring is 75 cm.

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SCORING MECHANICS, TIME BONUSES, PENALTIES & PROHIBITED EQUIPMENT

3. MATCH TIMING & TIME-BASED EXTRA POINTS

Matches consist of 3 rounds of **120 seconds (2 minutes)** each. Extra points are awarded for fast pulls and endurance dominance.

VICTORY / EVENT CONDITION	TIME FRAME / TRIGGER	BASE POINTS	SPEED BONUS	TOTAL AWARDED
Lightning Victory	Pull opponent out in 0 – 30 seconds	+100 Pts	+50 Extra Pts	150 Points
Rapid Victory	Pull opponent out in 31 – 60 seconds	+100 Pts	+25 Extra Pts	125 Points
Standard Victory	Pull opponent out in 61 – 120 seconds	+100 Pts	+10 Extra Pts	110 Points
Endurance Hold Win	Rope ring on your side at 120s timeout	+50 Pts	+20 Extra Pts	70 Points

4. POINT DEDUCTIONS & INFRACTIONS

INFRACTION TYPE	DESCRIPTION / TRIGGER	DEDUCTION / ACTION
False Start / Early Pull	Applying motor power before referee whistle	-15 Points (2nd offense = Round Forfeit)
Arena Surface Damage	Sharp metallic spikes/drills gouging floor mat	-30 Points + Immediate tool removal
Cable Entanglement (Wired)	Intentionally snagging or pulling opponent wire	-50 Points
Unauthorized Human Touch	Operator touching bot or rope during active match	Round Forfeiture (0 Pts for round)
Weight / Voltage Violation	Exceeding tolerance range (>3.15kg / >5.25kg)	Match Disqualification

5. STRICTLY PROHIBITED EQUIPMENT & HAZARDOUS SYSTEMS

BANNED HAZARDOUS SYSTEMS (AUTOMATIC DISQUALIFICATION)

- Fire & Thermal Equipment (STRICTLY PROHIBITED):** Flame throwers, spark generators, pyrotechnics, heat guns, or thermal cutters.
- Water & Liquid Systems (STRICTLY PROHIBITED):** Water cannons, liquid sprayers, oil/grease slicks, chemical solvents, or hydraulic fluids.
- Sticky & Adhesive Compounds:** Tapes, glues, suction pads, sticky gels, or chemical traction fixatives.
- Untethered Projectiles & Nets:** Harpoons, thrown weights, launched nets, or detachable mechanical traps.
- Signal Jammers & Frequency Blockers:** Active RF jammers, radio disrupters, or high-power EMP devices.

OFFICIAL TECHNICAL INSPECTION CHECKLIST

MANDATORY QUALIFICATION & SAFETY SCRUTINY

TECHNICAL & SAFETY COMPLIANCE VERIFICATION

STATUS	INSPECTION ITEM / PARAMETER	RESULT
☐	Category verified	PASS / FAIL
☐	Weight within category limit	PASS / FAIL
☐	30 × 30 × 30 cm envelope verified	PASS / FAIL
☐	12 V battery system verified	PASS / FAIL
☐	Battery securely mounted	PASS / FAIL
☐	Wiring insulated and secure	PASS / FAIL
☐	Stop / emergency control verified	PASS / FAIL
☐	Pull attachment secure	PASS / FAIL
☐	No fire equipment	PASS / FAIL
☐	No water equipment	PASS / FAIL
☐	No prohibited devices	PASS / FAIL
☐	Arena boundary and ±40 cm marks ready	PASS / FAIL
☐	Safety barrier / exclusion zone ready	PASS / FAIL
☐	Team briefed on STOP command	PASS / FAIL

INSPECTION NOTES & SCRUTINEER SIGN-OFF

INSPECTION NOTES / CORRECTIVE ACTION:

[Record required modifications, weight re-checks, or cable adjustments prior to final match clearance]

Inspector: _____

Team: _____

Date: _____

OFFICIAL JUDGE SCORE SHEET

PRINT ONE SHEET PER MATCH

MATCH SETUP & INFO

FIELD	ENTRY
Event / Venue	
Date	
Match Number	
Category	☐ Junior (Class 6-9) ☐ Senior (Class 10-12)
Team A	
Team B	
Start Time / Finish Time	Start: Finish:
Winner / Draw	

SCORING & POINTS CALCULATION

SCORING CRITERIA / EVENT	SELECTION	POINTS
Team A Standard Win	☐ Yes	+100
Team B Standard Win	☐ Yes	+100
Team A Lightning Bonus (0-30s)	☐ Yes	+50
Team B Lightning Bonus (0-30s)	☐ Yes	+50
Team A Quick Bonus (31-60s)	☐ Yes	+25
Team B Quick Bonus (31-60s)	☐ Yes	+25
Endurance Hold Bonus	Team: ☐	+20
Penalties / Deductions	Team A: Team B:	
FINAL SCORE	Team A: Team B:	

OFFICIAL RESULT & APPROVALS

OFFICIAL RESULT: ☐ TEAM A WIN ☐ TEAM B WIN ☐ DRAW ☐ ROUND FORFEIT ☐ DISQUALIFICATION

JUDGE NOTES / INCIDENT RECORD:

[Record warnings, surface damage, electrical violations, or false starts here]

Judge Signature: _____

Head Judge: _____

Safety Official: _____

Team Representative: _____

EVENT ORGANISER VERIFICATION & APPROVAL

Organiser Name: _____

Signature: _____

Date: _____